

Algebra 1
Unit 2
Lesson 1 Homework

Name _____
Date _____
Period _____

1. The product of powers law of exponents states to keep the then _____ the powers.

☐ add
☐ subtract
☐ multiply

☐ base
☐ coefficient
☐ exponent

and

2. Jeffrey simplified the algebraic expression and made an error. Find and fix the mistake.

$$\begin{aligned} &(-2x^2y^3)^4(3.5x^0y) \\ &(-2x^8y^{12})(3.5y) \\ &(-2 \cdot 3.5)(x^8)(y^{12} \cdot y) \\ &\quad -7.5x^8y^{13} \end{aligned}$$

3. Write a note to Jeffrey explaining the error he made in the previous question.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-4a^2b)(5ab^3)$

5. $(\frac{-1}{2}x^3y^0)(6x^2y^3)(-3y)$

$$6. \quad (5.6kj^8)(-3.25k^9)(\frac{1}{4}k^7j^{12})$$

$$7. \quad (-5mn^3p^5)^2$$

$$8. \quad (4.1c^{-4}d)(-2.2c^{10}d^0)(-c^{-2}d^3)$$

$$9. \quad (\frac{5}{8}r^{-2})^0(-7.6r^4s^5)^{-2}$$

$$10. \quad (-2y^3z)^5(6y^{10}z^8)^2$$